

Chapter 1: Mechanics & Motion

Kinematics, Newton's Laws of Motion, Work, Energy & Power

1.1 Kinematics: Describing Motion

Kinematics is the branch of classical mechanics that describes the motion of points, bodies (objects), and systems of bodies without consideration of the forces acting on them. Key parameters include displacement (Δx), velocity ($v = \frac{dx}{dt}$), and acceleration ($a = \frac{dv}{dt}$).

Equations of Uniformly Accelerated Motion:

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$

1.2 Newton's Laws of Motion & Dynamics

Dynamics incorporates the forces causing the motion. Sir Isaac Newton's three fundamental laws form the bedrock of classical mechanics:

- First Law (Inertia):** An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force.
- Second Law ($F = ma$):** The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object.
- Third Law (Action-Reaction):** For every action, there is an equal and opposite reaction.

1.3 Work, Energy & Conservation Principles

Work (W) is defined as the scalar product of force ($\vec{F}$) and displacement ($\vec{s}$): $W = F s \cos(\theta)$.

- Kinetic Energy (KE):** $KE = \frac{1}{2}mv^2$
- Potential Energy (PE):** Gravitational PE ($= mgh$), Elastic PE ($= \frac{1}{2}kx^2$)
- Work-Energy Theorem:** The work done by the net force on an object equals the change in its kinetic energy ($W = \Delta KE$).